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Rakshak evaluation report

Methodology, confusion matrix, per-language and per-type breakdowns, latency, policy check, system tests and limitations for the Rakshak scam-moment guardian

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Basis: data/runtime/eval_results.json (run of 17 October 2026, gemini-3.5-flash, n=50), scripts/eval.py, scripts/smoke_ws.py, scripts/smoke_live_audio.py, docs/EVAL_RESULTS.md

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> **Bottom line:** On 50 labelled synthetic calls (25 scam, 25 benign; Hindi, Hinglish, English, Singapore English, Kannada, Tamil; 10 hard negatives and 10 adversarial scams written to evade keyword detection), the gemini-3.5-flash coercion rubric separates every scam from every benign call at the 0.70 hold threshold (precision 1.0, recall 1.0, F1 1.0). The deterministic fusion policy makes the labelled payment decision on all 39 payment attempts (accuracy 1.0); the first run missed one, an ambiguous job-scam call that has since moved to a hold because the payee scoring was refined (section 2.6). Per-signal accuracy over the eight coercion signals is 0.932. The mean rubric latency was 3.1 s (median 2.0 s) over the 50-call evaluation run, and 2.4 to 3 s for a single uncontended request. Three system tests exercise the live audio path end to end on the deployed service. The dataset is self-authored and small, and a perfect score on it is not a field accuracy; the next step with a bank partner is a human-labelled corpus of real calls.

1. What is measured and how

1.1 The two decisions under test

Rakshak makes two decisions, and the harness (`scripts/eval.py`) scores both:

1. **Call-level detection.** The transcript up to the payment attempt (or the whole call when there is none) is sent to the coercion rubric on `gemini-3.5-flash` with a strict JSON schema. The rubric returns a language, a scam type (10 classes), a stage, eight boolean coercion signals, a risk score from 0 to 1 and reasons in English and in the customer's language. A call is counted as "detected" when its risk score is at or above the hold threshold of 0.70, which is the same threshold the product uses.
2. **Payment-level decision.** For every call that contains a payment attempt, the fusion policy (`app/policy.py`) combines the rubric's risk score and signals with the payee's mule-risk row and the customer profile and returns one of proceed, warn, hold or hold_and_alert. The decision is correct when a scam call gets hold or hold_and_alert and a benign call gets proceed or warn. This is the deterministic fast path that executes in under 100 ms at the payment moment; the ADK agent, which runs in parallel and can only make the action stricter, is not part of the evaluation loop.

A third measurement, per-signal accuracy, compares each of the eight predicted signals against the labelled expectation for every call (8 x 50 = 400 comparisons).

1.2 Dataset design

The dataset is `data/eval/calls.json`: 50 synthetic calls written for this project, each with an id, label, scam type, language, a victim profile (age band, preferred language, trusted contact), the turn-by-turn transcript with speaker labels (Hindi and other scripts carry romanised and English renderings for review), the eight expected signals, an expected risk band, an optional payment attempt (payee VPA, amount, the turn index at which it happens) and a note explaining what the call is meant to test.

Block	Ids	Count	Purpose
Base scams	call_001 to call_015	15 scam	Digital arrest, investment, job task, KYC/OTP, lottery, family emergency, Singapore official impersonation; the patterns described in I4C, RBI, SPF and MAS advisories
Base benign	call_016 to call_030	15 benign	Everyday calls that mention money, deadlines or officials without coercion
Hard negatives	call_031 to call_040	10 benign	Benign calls that share the surface vocabulary of scams: a genuine bank fraud desk saying "blocked" and "OTP", a real police station about a found wallet, a hospital deposit for same-day surgery, a Kannada son abroad needing a visa fee by tomorrow, an insurance premium "due today", school fees with a late fee, a harsh Tamil landlord threatening a notice to vacate, a courier asking for a small COD amount, a SEBI-registered adviser discussing 30 percent returns, a friend asking to be repaid

Block	Ids	Count	Purpose
Adversarial scams	call_041 to call_050	10 scam	Scams rewritten to defeat keyword rules: a digital arrest that never says police, arrest, OTP or safe account ("department", "case file", "monitoring account"); a family emergency with no authority and a soft "he's okay" framing; a slow-burn investment pitch that disowns the word "guarantee"; a refund scam that asks for a "confirmation amount"; a job scam framed as a refundable training kit; a KYC scam that calls the OTP a "matching code with nothing to do with money"; a Kannada loan-app scam citing "RBI rules" without claiming to be RBI; a Tamil prize scam that says "this isn't a lottery"; a Singapore parcel scam with "holding account for safekeeping"; a UPI customer-care scam that promises never to ask for a PIN and routes the balance through a "temporary migration wallet"

Languages: Hindi (15 calls), Hinglish (15), English (12), Singapore English (4), Kannada (2), Tamil (2). Thirty-nine of the fifty calls include a payment attempt; the eleven without one are benign calls with no transfer, which still count for detection and signal accuracy.

Payee rows for the payment attempts come from the same 64-payee synthetic table the product uses (`data/payees.json`, bands low, medium, high and critical). Several adversarial scams deliberately pay a new high-risk or critical row so that the payee side of the fusion is tested in cases where the transcript alone is ambiguous. The victim profile used by the harness sets a typical transfer of 5,000 rupees so the "amount more than 5x usual" bump is exercised.

1.3 Run conditions

The reported run is `data/runtime/eval_results.json` written on 17 October 2026 (run_at 1792261796, a re-run after the payee-scoring change described in section 2.6) with model `gemini-3.5-flash`, hold threshold 0.70, concurrency 4. The same results are rendered in `docs/EVAL_RESULTS.md`. An offline keyword stub (`--stub`) exists for CI and is not reported here.

2. Results

2.1 Call-level detection

Confusion matrix at the hold threshold 0.70 (rows are labels, columns are predictions):

	Predicted scam (risk >= 0.70)	Predicted benign (risk < 0.70)
Scam (25)	25	0
Benign (25)	0	25

Metric	Value
Precision	1.000
Recall	1.000
F1	1.000
Lowest risk score on a scam call	0.75 (call_045, the ambiguous job-scam call)
Highest risk score on a benign call	0.15 (call_033, the hospital deposit)
Scam-type classification on scam calls	25 of 25 match the label

The margin matters more than the headline: the lowest scam score (0.75) sits just above the threshold, while the highest benign score (0.15) is far below the warn threshold of 0.45. The hard negatives, which were written to look like scams on vocabulary alone, all scored 0.15 or lower, including the genuine fraud-desk call that says "blocked", "OTP" and "safer" (call_031, 0.00) and the Tamil landlord threatening a notice to vacate (call_037, 0.00).

2.2 Payment-level decision (fusion policy)

Metric	Value
Payment attempts	39 (25 scam, 14 benign)
Correct decisions	39
Payment-decision accuracy	1.000
Benign payments that got warn instead of proceed	0
Scam payments that got a hold	25 of 25, all hold_and_alert (every evaluation profile has a trusted contact)

No payment decision is wrong in this run. The case to watch is call_009, a Hinglish job-task scam (risk 0.85, predicted type job_task) paying 2,000 rupees to `tasks.hr.payout@rakshakdemo`. The ledger-derived lookup placed the payee in the medium band at run time, the recorded fused score is 0.775, and the action is hold_and_alert. In the first run this call scored 0.67 against a low-band payee and got a warn, which was the one miss. Its margin over the 0.70 hold threshold is 0.075 and it is the lowest fused score among the 25 scam payments (the next lowest is call_045 at 0.985); section 4 shows how much of that margin depends on the payee term and on the `safe_account` flag.

2.3 Per-signal accuracy

Over 400 signal comparisons (8 signals x 50 calls), 373 are correct (0.932).

Signal	Correct of 50	False positives	False negatives	Accuracy
secrecy	50	0	0	1.00
otp_or_credential_request	49	0	1	0.98

Signal	Correct of 50	False positives	False negatives	Accuracy
threat_of_arrest_or_harm	49	1	0	0.98
authority_impersonation	47	3	0	0.94
investment_pressure	47	0	3	0.94
remote_app_install	46	3	1	0.92
urgency	43	0	7	0.86
safe_account	42	8	0	0.84

Two signals account for most of the disagreement. `safe_account` is over-predicted (8 false positives): the model marks the flag on scams whose payment destination is a "monitoring account", "holding account" or "migration wallet" even where the label reserved the flag for an explicit "safe account" demand. This over-prediction only ever occurs on scam calls, so it adds a +0.10 bump that pushes the fused score the right way; it does not affect any benign call, and it is decisive in one case, call_009 (section 4). `urgency` is under-predicted (7 false negatives): the model declines to flag urgency on calls where the deadline is implied rather than stated. Both are labelling-convention disagreements rather than detection failures, and neither changed a call-level outcome; the one payment-level outcome they touch is call_009, discussed in section 4.

2.4 Per-language breakdown

Computed from `data/runtime/eval_results.json`.

Language	Calls	Scam / benign	Detection correct	Payment decisions correct	Signal accuracy	Mean rubric latency
Hindi (hi)	15	8 / 7	15 / 15	11 / 11	0.917	2.3 s
Hinglish	15	8 / 7	15 / 15	13 / 13	0.925	2.1 s
English (en)	12	5 / 7	12 / 12	8 / 8	0.948	4.1 s
Singapore English (en-sg)	4	2 / 2	4 / 4	3 / 3	1.000	8.3 s
Kannada (kn)	2	1 / 1	2 / 2	2 / 2	0.938	2.3 s
Tamil (ta)	2	1 / 1	2 / 2	2 / 2	0.875	2.2 s

Detection and payment decisions are perfect in every language. Signal accuracy is lowest in Tamil (two calls) and in Hindi and Hinglish, where the `safe_account` and `urgency` conventions above are concentrated. The English and Singapore English latency means are each lifted by one call that took about 28 s (call_026 and call_049); see section 3. Kannada and Tamil are covered by two calls each (one scam, one benign); this is a smoke test of the rubric's reading of those scripts, not a measurement of its accuracy in them.

2.5 Per-scam-type breakdown

Scam type	Calls	Detection correct	Payment decisions correct	Signal accuracy	Mean rubric latency
digital_arrest	6	6 / 6	6 / 6	0.958	2.4 s
investment	4	4 / 4	4 / 4	0.812	2.2 s
kyc_otp	4	4 / 4	4 / 4	0.906	2.3 s
job_task	3	3 / 3	3 / 3	0.750	2.1 s
family_emergency	2	2 / 2	2 / 2	0.938	2.0 s
lottery	2	2 / 2	2 / 2	0.875	3.1 s
sg_official_impersonation	2	2 / 2	2 / 2	1.000	15.1 s

Scam type	Calls	Detection correct	Payment decisions correct	Signal accuracy	Mean rubric latency
loan_app	1	1 / 1	1 / 1	1.000	1.7 s
other (refund scam)	1	1 / 1	1 / 1	1.000	1.8 s
benign (all)	25	25 / 25	14 / 14	0.965	3.0 s

The adversarial block (call_041 to call_050) scored between 0.75 and 1.00, all above the threshold, with the predicted scam type matching the label in every case, including the refund scam labelled `other`. The two slow cells are not about content: sg_official_impersonation (15.1 s) is call_049 at 28.4 s, and the benign mean (3.0 s) includes call_026 at 27.9 s. The job_task type has the lowest signal accuracy (0.750) because on all three calls the label expects `investment_pressure`, which the model does not flag, and the model flags `safe_account`, which the label does not expect.

2.6 What changed since the first run

The first run of this set reported payment-decision accuracy 0.974 (38 of 39); this run reports 1.0 (39 of 39). The only call that moved is call_009. Its rubric output is the same as before (risk 0.85, type job_task, `safe_account` flagged); what changed is the payee side. After the first run the ledger scorer got a graded fan-in step (0.30 for 8 or more distinct payers in 48 hours, 0.15 for 4 to 7) and the ledger stream was made steady-state per payee profile, so the payee term no longer drifts with uptime. The payee behind call_009 now reads as medium risk, which lifts the fused score from 0.67 to 0.775 and turns the warn into a hold. This is a refinement of the payee scoring, not a change to the labelled data: the 50 calls, their labels and their expected signals are unchanged, and the call-level numbers (precision, recall, F1 1.0) and the signal accuracy (0.932) are the same as in the first run. Two cautions apply. The rubric model is non-deterministic, so a further re-run can differ by a call or two, in either direction, in risk scores, signal flags and therefore possibly in a borderline payment decision such as this one. And 39 payment decisions from a self-authored set is a small sample: moving from 38 to 39 correct is one call, and should be read as the policy behaving sensibly on the cases we wrote, not as a measured accuracy of 100 percent in the field.

3. Latency

Measurement	Value	Condition
Rubric, single request	2.4 to 3 s	Uncontended, `thinking_level=minimal`, measured during development and in the smoke tests
Rubric, evaluation run	3.1 s mean (median 2.0 s, minimum 1.3 s, maximum 28.4 s)	50 calls at concurrency 4 against the API; two calls (call_026 and call_049) took about 28 s each and the other 48 averaged 2.1 s, so the tail looks like API retry or rate-limit backoff rather than model time (not confirmed per call)
Fusion policy	under 100 ms	Pure arithmetic in the request; the case is stored with `latency_ms: 0`
Hold event on the phone after Pay	20 ms to 1.2 s	Measured on the public URL by `scripts/smoke_ws.py` (decision event timestamp minus the Pay request)
Voice barge-in after the hold event	about 100 ms	When the clip was prewarmed (startup, pay-screen open, or replay start); otherwise after TTS synthesis of about 6 s, during which the hold screen is already up
ADK agent decision	11 to 17 s	`gemini-3.8-flash` at `thinking_level=low`, two agents, two to three tool calls; off the critical path

The design consequence: the rubric is scheduled with a 1.2 s debounce and at least 3 s between calls so that a 2 to 3 s model never falls behind a conversation, and the payment decision does not wait for any model at all.

4. The payment-decision policy check

The fusion formula is `score = 0.65 x call_risk + 0.35 x payee_risk`, plus +0.10 for a safe-account or credential demand, +0.05 for authority impersonation combined with a threat, +0.10 for a high or critical payee during a coercive call, +0.05 for amounts of 50,000 rupees or more and +0.05 for amounts above five times the customer's usual transfer; warn at 0.45, hold at 0.70, hold_and_alert when the profile has a trusted contact.

Checking the two boundary cases in the set:

- **call_009 (job scam, medium-band payee at run time, 2,000 rupees):** 0.65 x 0.85 gives 0.5525; the rubric flagged `safe_account` (the caller routed the fee through a "payout" account), which adds 0.10 and gives 0.6525; the recorded fused score is 0.775, so the payee term is 0.1225, which is 0.35 x a ledger-derived payee score of 0.35 (exactly the lower edge of the medium band). No authority-plus-threat bump, no high-risk-payee bump (the band is medium, not high or critical) and no amount bumps. Result 0.775, hold_and_alert, 0.075 over the hold threshold. The outcome is correct and the margin is thin, in two ways. Without the payee term the call would score 0.6525 and get a warn; the payee needs a score of about 0.14 or more to tip it. And the `safe_account` flag is a false positive against the label (the label expects `investment_pressure` and no safe-account demand); without that +0.10 the same call would score 0.675 and again be a warn. So the hold here is right but rests on a medium payee and on a signal the label does not endorse. That is acceptable for a rule whose job is to stop a likely scam, and it is why the product keeps the ADK agent beside the policy: it sees the same evidence and may upgrade a borderline warn, and the policy floor guarantees it cannot go the other way. In the first run the payee scored lower (low band, fused 0.67) and this call was a warn; the only reason to lower the 0.70 hold threshold for it would be to hold every small transfer to a clean payee whenever a call is merely suspicious, which is the friction that makes customers switch a guardian off, so the threshold was left alone.
- **call_033 (hospital deposit, benign, established hospital VPA):** rubric 0.15, low-band payee, amount 60,000 rupees, so both amount bumps apply (+0.05 for 50,000 or more, +0.05 for more than five times the 5,000-rupee typical transfer). The recorded fused score is 0.302: 0.65 x 0.15 gives 0.0975, the amount bumps add 0.10, and the remaining 0.105 is the payee term (a ledger-derived payee score of about 0.30, still in the low band, which ends at 0.35). That is the highest fused score of any benign payment and it is 0.148 under the warn threshold of 0.45, so the action is proceed. The other hard negatives with urgency and money (call_034 visa fee of 42,000, fused 0.188; call_036 school fees of 21,500, 0.138; call_037 rent of 18,000, 0.105) resolve to proceed for the same reason: with call risk near zero, no payee bump and no safe-account bump, the amount bumps (at most 0.10) plus a low-band payee term stay well under the warn threshold. Note that a ledger-derived payee score near 0.30 for a large legitimate hospital is the graded fan-in step working as designed (many payers alone is normal for a hospital), and it leaves this case about 0.05 from the medium band, which is the next thing to calibrate against real ledgers.

The policy is unit-tested in `tests/test_policy_and_audio.py` and is the floor for the agent: `final = max(agent_action, policy_action)` in `app/payment.py`, so nothing the agent says can turn a policy hold into a proceed.

5. System tests on the deployed service

These are not scored like the dataset; each is a pass/fail check of a path the dataset cannot exercise.

Test	What it does	Result
`scripts/smoke_ws.py <url>`	Creates a text-mode session, posts the four-line Hindi CBI script, pays 2 lakh to `cbi.verification@rakshakdemo`, measures time to the `decision` event and the first private `rakshak_audio` message, waits for the ADK `case_update`, then runs a typed turn through the post-hold `gemini-3.8-live` conversation socket	Passed on the public URL: hold event 20 ms to 1.2 s after Pay, voice about 100 ms when prewarmed, agent update `decided_by adk_agent` within the 60 s window, Live conversation answered
`scripts/smoke_live_audio.py <url>`	Synthesises three Hindi scam sentences as caller audio, rings the session over `/ws/call/{id}/caller`, accepts as the user, streams the audio at real-time pace in 100 ms chunks with 1.5 s pauses, and reads the events socket	Passed: 3 correct transcript turns from `gemini-3.5-transcribe-live`, risk updates 0.9 to 1.0, whisper fired, audio relayed to the user leg
Two-way relay	Two browsers on the same session, each with a microphone, speaking in turn	Both parties received each other's audio; each leg transcribed on its own Live session with the correct speaker label
`uv run pytest`	16 offline tests with `RAKSHAK_FAKE_LLM=1`: the API flow, the policy, audio helpers, the ledger scorer and stream (including a regression test that simulates three days of uptime and checks that payee scores do not drift)	Passed (16 passed)

6. Limitations

- **Self-authored data.** All 50 calls were written by the team to resemble patterns in public advisories. The adversarial block was written after the base set, by the same people, with the rubric prompt in front of them; it raises the bar but does not remove the authorship bias. A perfect score on this set is evidence that the rubric reads coercion rather than keywords, not evidence of field accuracy.
- **Small languages.** Kannada and Tamil have two calls each. Hindi output quality was checked by a Hindi speaker; Kannada and Tamil outputs (rubric reasons and TTS) have not been reviewed by native speakers.
- **Transcript, not audio.** The dataset evaluates the rubric on clean text. Transcription errors from real speech (accents, line noise, code-switching mid-sentence) are only covered by the single live-audio smoke test. The live transcriber emits per utterance, so a long sentence can land as two turns.
- **Two runs, one model, non-deterministic.** The numbers are from the latest of two evaluation runs of `gemini-3.5-flash`; the rubric is non-deterministic, so a re-run can differ by a call or two, and model updates can move scores. The harness writes a timestamped result on every run so a later run can be compared. The payee-scoring refinement between the two runs was made after seeing the first run's miss, so the 1.0 payment accuracy should not be read as untouched held-out performance.
- **The agent is not scored.** The ADK pipeline's decisions, tool calls and user messages are captured on every case but were not evaluated against labels; the policy floor means the agent cannot lower a score in the dataset sense, but its upgrade and message quality are unmeasured.
- **Synthetic payees and ledger.** Payee rows are synthetic; the ledger stream reproduces the profile shapes and, since the refinement, is steady-state per payee profile (mule accounts receive new victims at their own rate, shops and individuals get repeat customers, about 150 transactions a minute, re-seeded every 6 hours), so payee scores no longer depend on how long the service has been up, though exact fused scores may still vary slightly after a re-seed. The fusion's payee term is therefore tested on designed, not observed, distributions.
- **No override rate.** The product's most important field metric, how often customers override a hold and whether those overrides were right, cannot exist without users.

7. What a bank would do next

1. **Real call corpus, human-labelled.** Take consented recordings from the bank's own in-app calls or a telco partner's fork, transcribe them with the same `gemini-3.5-transcribe-live` path, and have two fraud analysts label scam type, stage and the eight signals with adjudication of disagreements. Target 500 calls with at least 100 confirmed scams and 100 hard negatives from the bank's complaints desk (genuine fraud-desk calls, collections calls, hospital and school payments).
2. **Audio-in evaluation.** Score the rubric on the transcriber's output rather than clean text, so transcription errors are inside the measurement, and report word error rate per language alongside.
3. **Threshold calibration per segment.** Fit the warn and hold thresholds (already per-profile fields) to the bank's loss and friction targets per customer segment, and publish the false-positive rate at the chosen operating point rather than at a single 0.70.
4. **Override and outcome tracking.** Log every override with its reason and follow the payment to its outcome (complaint filed, chargeback, nothing) so precision is measured on consequences, not labels.
5. **Agent evaluation.** Score the ADK decision agent's action and the user message against the analysts' labels, and the authority check against the advisory table, so the agent's upgrades can be trusted and its message quality improved in each language.
6. **Ledger features from real data.** Replace the synthetic ledger with `sql/payee_risk.sql` on the bank's own transactions and the DoT FRI, I4C and KYC feeds, and re-run the payment-level check with real payee rows.
7. **Native-speaker review.** Kannada and Tamil rubric reasons and TTS clips reviewed and corrected before any customer in those languages sees them.